

Insurance Fraud Study

Business-oriented analysis of fraud patterns, operational scoring and counterfactual decision support

Prepared for management audiences. The objective is to explain where fraud risk concentrates, how the scoring tool should be used in operations, and what corrective levers the counterfactual layer surfaces.

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Executive Summary

- The historical portfolio contains 54,248 claims with an observed fraud rate of 7.0%. Fraud is therefore a minority event, which makes prioritization more valuable than blanket review.
- The average annual fraud loss observed in the historical portfolio is about 8,505,676 €. This is the right figure to communicate to management because it translates fraud into an annual business cost.
- Fraud is not spread evenly across the portfolio. Bodily injury and fire claims are the most exposed categories, and suspicious providers materially amplify risk.
- The deployed scoring setup is designed as a triage tool. On the historical holdout, the riskiest 1% of files reaches 77.8% precision, which is a strong signal for prioritizing manual review queues.
- On the first 100 production files, only 4.0% crossed the fraud decision threshold. This suggests the tool is selective rather than alarmist, which is preferable for operational adoption.
- The counterfactual layer adds managerial value: it does not only say that a file is risky, it shows which levers would most plausibly bring the file back below the decision boundary.

1. Business Context And Study Objective

The aim of this study is to equip the company with a practical fraud decision-support workflow. A client file arrives with CRM information, a PDF and an image; the information is extracted and stored in PostgreSQL; analysts explore the portfolio in SQL; a scoring model ranks incoming files by risk; and a counterfactual layer explains what would need to change for a suspicious claim to become acceptable.

For a management audience, the key question is simple: *where should teams focus first, what recurring fraud patterns are visible in the portfolio, and how can the tool support decisions without replacing human review?*

2. Simplified Methodology

- Step 1. Consolidate each claim package into structured fields and load the results into PostgreSQL.
- Step 2. Use SQL and exploratory analysis to understand portfolio shape, fraud concentration zones and suspicious operational patterns.
- Step 3. Build a scoring model on historical data only, then tune it as a prioritization tool rather than a blanket rejection machine.
- Step 4. Apply the model to new production files after the development boundary, and complement the score with counterfactual explanations.

This sequencing matters because the study does not jump directly to machine learning. The analysis starts with SQL and business profiling, which is the right way to understand data quality, recurring behaviors and operational segments before training a model.

3. Portfolio Reading: What The Historical Data Tells Us

Indicator	Value
Historical files	54,248
Observed fraud rate	7.0%
Average annual fraud loss	8,505,676 €
Average claim amount	6,990 €
Median claim amount	4,173 €
Average fraud amount	15,644 €
Average non-fraud amount	6,337 €
Average customer age	51.6 years

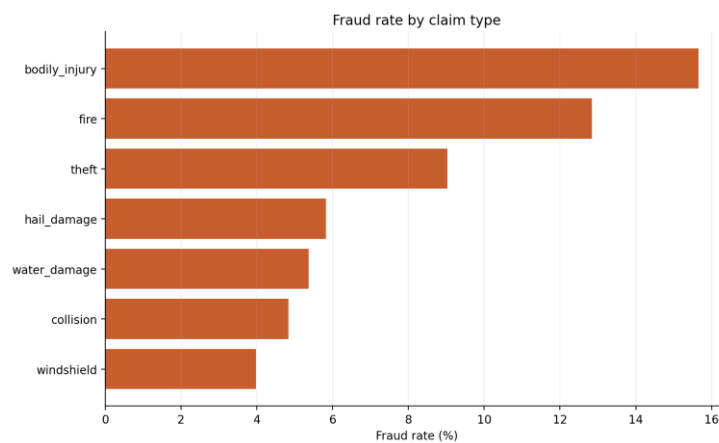
The first signal is severity. Fraudulent files are, on average, far more expensive than standard files. This justifies a risk-based review process because the financial downside of missing a suspicious claim is not uniform across the portfolio. Across the observed historical years, the average annual fraud loss is about 8,505,676 €, which gives the business a concrete annual cost reference.

Main findings from the exploratory analysis:

- Suspicious providers show a fraud rate of 12.0% versus 5.4% for the rest of the network.
- High-amount claims reach a fraud rate of 49.9%, compared with 5.8% outside the high-amount band.
- Claims filed shortly after a previous claim reach 13.5% fraud rate, versus 5.9% otherwise.
- Newer policies are more exposed: files with short policy tenure show 9.9% fraud rate, versus 6.7% for more mature contracts.

3.1 Claim Types With The Highest Exposure

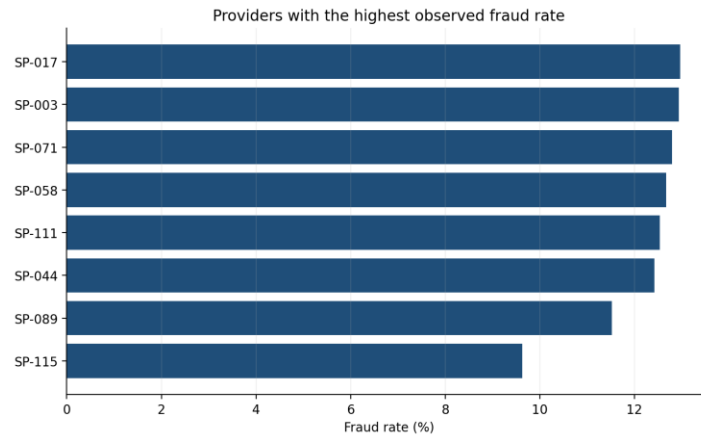
Fraud is concentrated in a few claim families, which is useful for operations because training, escalation rules and review playbooks can be specialized.



Claim type	Files	Fraud rate	Avg amount
Bodily injury	6,011	15.7%	18,214
fire	3,021	12.8%	16,395
theft	6,503	9.0%	11,628
Hail damage	6,524	5.8%	2,512
Water damage	6,990	5.4%	4,770
collision	15,456	4.8%	5,212
windshield	9,743	4.0%	1,463

Bodily injury and fire stand out clearly. These categories combine higher average amounts with higher fraud incidence, which makes them natural candidates for stronger review rules and more detailed documentation requirements.

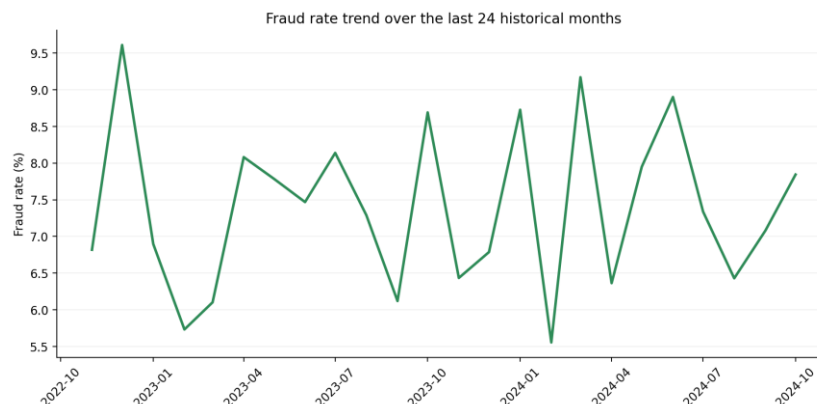
3.2 Provider Network Risk



Provider	Files	Fraud rate	Avg amount
SP-017	1,673	13.0%	7,236
SP-003	1,492	12.9%	7,428
SP-071	1,751	12.8%	7,443
SP-058	1,729	12.7%	7,062
SP-111	1,508	12.5%	7,110
SP-044	1,771	12.4%	6,709
SP-089	1,666	11.5%	7,047
SP-115	270	9.6%	7,103

The provider effect is strong enough to deserve explicit operational follow-up. This does not prove misconduct on its own, but it gives management a short list of partner relationships that merit audit, controls reinforcement or pricing review.

3.3 Recent Trend



The monthly fraud trend does not point to a runaway structural deterioration, but it does show periodic spikes. This supports a monitoring setup where alerts focus on deviations by month, claim type and provider cluster, rather than on raw portfolio volume alone.

4. From Analysis To Modeling

The modeling stage was intentionally positioned after the SQL and exploratory phase. In practice, the analysis identified the most useful levers such as claim severity, suspicious providers, claim recency, short policy age and category-specific context. The model then learns how these signals combine.

For management, the key point is not the technical family of the model. The important point is the operating mode, cause the tool is configured to be selective. It favors precision on the high-risk queue so that analysts spend time on a shorter list of files with stronger fraud concentration.

Review bucket	Files reviewed	Observed precision	Observed recall
Top 1% highest-risk files	108	77.8%	11.0%
Top 5% highest-risk files	542	43.2%	30.7%
Top 10% highest-risk files	1085	30.2%	43.1%

This table is the most business-friendly way to read the model. If the company wants a very focused review queue, the top 1% is already highly concentrated. If the company wants broader coverage, the top 5% or top 10% expands the queue while keeping fraud concentration above baseline.

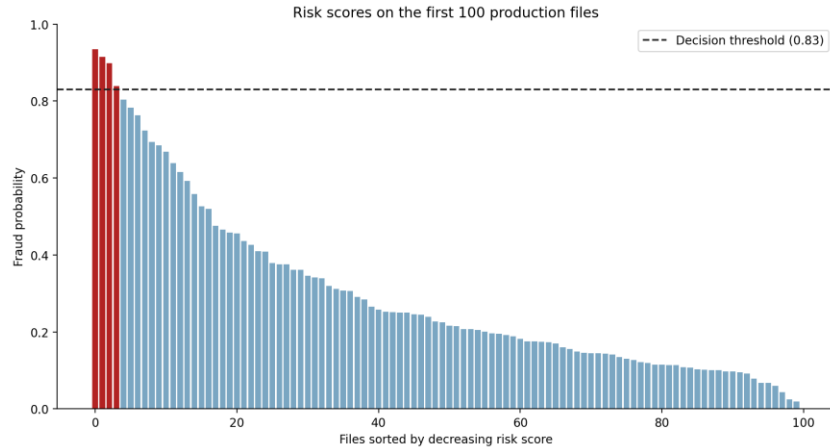
Signals most frequently picked up by the model:

- High Amount Flag (0.104)
- Suspicious Provider Flag (0.046)
- Claim Amount (0.016)
- Time Since Last Claim Days Filled (0.009)
- Recent Claim Flag (0.009)

5. Focus On The First 100 Production Files

To make the deployment story concrete, the study reviews the first 100 production files scored after the development freeze. This is useful because it translates an abstract model into an operational intake queue.

Indicator	Value
Files reviewed	100
Average risk score	0.298
Share above 0.50 score	17.0%
Share above deployment threshold	4.0%
Share routed through suspicious providers	34.0%
Share in high-amount band	4.0%



The first 100 files suggest a healthy deployment posture. The tool does not flag large volumes indiscriminately. Instead, it isolates a narrow subset of files that should be escalated to manual review, while preserving analyst attention for the most informative cases.

Claim ID	Date	Type	Location	Provider	Amount	Score	Decision
CLM-03452-003	2024-10-07	Bodily injury	Rennes	SP-098	52,508	0.935	fraud
CLM-02612-008	2024-10-08	Bodily injury	Metz	SP-016	39,795	0.915	fraud
CLM-08154-010	2024-10-07	Bodily injury	Saint-Étienne	SP-007	40,223	0.898	fraud
CLM-05314-013	2024-10-04	windshield	Nîmes	SP-071	1,658	0.839	fraud
CLM-06755-005	2024-10-06	Theft	Villeurbanne	SP-058	23,076	0.804	Non-fraud
CLM-02424-011	2024-10-05	collision	Dijon	SP-003	6,835	0.782	Non-fraud
CLM-03043-006	2024-10-08	Bodily injury	Caen	SP-074	29,962	0.764	Non-fraud
CLM-00858-010	2024-10-08	windshield	Paris	SP-003	1,838	0.723	Non-fraud
CLM-01797-006	2024-10-03	collision	Marseille	SP-012	3,716	0.694	Non-fraud
CLM-02191-005	2024-10-05	Water damage	Rennes	SP-098	8,672	0.685	Non-fraud

In this initial production sample, the riskiest files are not random. They often combine a costly claim profile with either a suspicious provider or a claim type that is historically overexposed to fraud.

6. Counterfactual Reading: What Would Need To Change?

A counterfactual explanation answers a management-friendly question: *If a claim is considered too risky, what would need to change for the claim to fall back below the decision threshold?* This is useful for auditability because it turns a score into an interpretable discussion.

Example CLM-03452-003

- Original file: bodily injury claim in Rennes routed through provider SP-098, amount 52,508, score 0.935.
- ✓ **Suggested change:** claim amount: 52507.91 -> 36755.54. The score would move from 0.935 to 0.445.

Example CLM-02612-008

- Original file: bodily injury claim in Metz routed through provider SP-016, amount 39,795, score 0.915.
- ✓ **Suggested change:** claim amount: 39794.96 -> 19897.48. The score would move from 0.915 to 0.351.

Example CLM-08154-010

- Original file: bodily injury claim in Saint-Étienne routed through provider SP-007, amount 40,223, score 0.898.
- ✓ **Suggested change:** claim amount: 40223.23 -> 20111.62. The score would move from 0.898 to 0.266.

In the current production sample, the first counterfactuals mainly point to provider changes. That is a meaningful business message: **the tool is not only reacting to raw claim amount, it is also recognizing that some operational counterparties materially change the risk level.**

7. Management Interpretation

- The main operational opportunity is prioritization. The tool can focus analyst effort on a much smaller set of claims with stronger fraud concentration.
- Provider governance matters. Several counterparties deserve specific review because their fraud exposure is materially above portfolio baseline.
- Claim-type playbooks should be differentiated. Bodily injury and fire deserve tighter controls than windshield or standard collision files.
- Counterfactual explanations make model outputs discussable. They help teams understand whether a file is risky because of severity, recency, provider choice or a combination of signals.
- The scoring layer should support human review, not replace it. This is especially important for relationship-sensitive cases and high-severity claims.

8. Recommended Next Actions

- Adopt a tiered review queue based on risk bands rather than a binary accept or reject process.
- Audit the top suspicious providers and tighten supporting-document requirements on the most exposed claim types.
- Track monthly fraud concentration by provider, claim type and location to detect drift early.
- Use counterfactual examples in analyst training so teams understand the concrete levers behind high-risk alerts.
- Extend the explanation layer with richer local interpretability tools if the company wants case-by-case governance evidence.

9. Limits Of The Study

- The dataset is synthetic, so portfolio effects should be read as a realistic simulation rather than as a live insurer benchmark.
- The extraction layer is local and rule-based; a production OCR and document understanding stack would enrich the intake process.
- Counterfactuals are heuristic: they are decision-support scenarios, not legal or causal proof.

Appendix: Technical Snapshot

For completeness, the current deployment champion is an XGBoost model tuned for precision with a minimum recall constraint. This appendix is intentionally brief because the main body of the report focuses on business interpretation.

Metric	Value
ROC AUC	0.765
Precision	0.525
Recall	0.247
F1	0.336
Decision threshold	0.83
Champion classifier	XGBoost
Imbalance strategy	oversample